

Nidita Roy

MSc Researcher in Computer Science | Medical Image Analysis | Class-Incremental Learning

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Education

MSc in Computer Science, University of Victoria, Canada

2025–Present

GPA: **8.50/9.00** | Supervisor: Dr. Ashery Mbilinyi

- Current thesis direction: *Class-Incremental Learning with Modality-Aware Representation Primitives*; research toward the thesis objectives is ongoing.
- **Coursework completed:** Systems for Massive Datasets (97%, A+), Data Mining (93%, A+), Deep Learning for Medical Image Analysis (92%, A+), Online Learning (81%, A-), and CSC 595.
- Portfolio: nidita.github.io

BSc in Computer Science and Engineering, Chittagong University of Engineering & Technology (CUET), Bangladesh

2019–2024

CGPA: **3.68/4.00** (First Class)

- Thesis: *Spinal Disease Detection via Deep Multimodal Fusion of X-ray Imagery and Demographic Features*.

Research Experience

Class-Incremental Learning with Modality-Aware Representation Primitives

Ongoing MSc Research

- Currently developing modality-aware representation approaches for sequential medical-imaging tasks across CT and MRI as part of my MSc thesis objectives.
- Working on task-conditioned CT representations that can adapt to clinically meaningful intensity characteristics while learning sequential tasks.
- Currently designing MRI pulse-sequence-aware representations for T1-weighted, T2-weighted, FLAIR, and DWI imaging to investigate how acquisition-specific information can be incorporated into continual-learning models.
- Assessing datasets including BraTS, AOMIC ID1000, and ADNI for their suitability for sequence-aware and modality-aware MRI experiments.
- Working toward integrating the proposed representation mechanisms into class-incremental learning pipelines and comparing them with standard continual-learning approaches.
- Planned evaluation focuses on knowledge retention, forward transfer, and performance across sequential medical-imaging tasks.

What Limits Continual Learning for Abdominal-Trauma CT Detection?

MICAD 2026

Co-lead author and corresponding author

- Diagnostic study separating catastrophic forgetting from representation, label, and task-learnability bottlenecks using the RSNA 2023 Abdominal Trauma Detection dataset.
- Evaluated fine-tuning, EWC, Replay, EWC+Replay, and LwF under class-incremental and domain-incremental settings.
- Contributed to experimental design, implementation, analysis, and manuscript writing.

Spinal Disease Detection via Deep Multimodal Fusion of X-ray Imagery and Demographic Features

BSc Thesis

Lead author

- Developed a multimodal deep-learning approach combining spinal X-rays with demographic information for eight-condition classification, followed by segmentation.
- Published on IEEE Xplore in 2026.

Patient Survival Prediction Using Actigraphy and Clinical Information

Cancers 2023

Research contributor

- Contributed to methodology, statistical analysis, and original-draft writing for a multimodal survival-prediction study integrating wearable actigraphy and clinical information.
- The deep-learning model achieved 92.4% accuracy and was compared with established prognostic tools including KPS and PPI.

Deep Learning for Wound Tissue Segmentation: A Comprehensive Evaluation Using a Novel Dataset

Preprint 2025

Research contributor

- Contributed to **data creation**, literature review, and implementation for a novel wound-tissue dataset.
- Evaluated full-image, patch-based, superpixel-based, segmentation, and classification formulations across a broad model benchmark.

Enhancing Offensive Bengali Social Media Meme Detection

ICCIT 2023

Research contributor

- Developed a weighted multimodal ensemble for offensive Bengali meme classification and contributed to dataset creation and modeling.
- Worked with the Multi-modal Offensive Memes Bengali Dataset (MoMBD), containing 3,662 memes.

Publications

1. L. E. Ismaila[†], **Nidita Roy**^{†,*}, S. Amin, C. Maroa, and A. Mbilinyi. “What Limits Continual Learning for Abdominal-Trauma CT Detection? A Diagnostic Study of Forgetting vs. Representation Bottlenecks.” Accepted at the **7th International Conference on Medical Imaging and Computer-Aided Diagnosis (MICAD 2026)**. [†]Equal contribution; ^{*}Corresponding author. [Paper] [Code]

2. **Nidita Roy** et al. “Spinal Disease Detection via Deep Multimodal Fusion of X-ray Imagery and Demographic Features.” **PECCII 2026**; published on IEEE Xplore. [IEEE Xplore]

3. M. A. Kabir, **Nidita Roy**, M. E. Hossain, J. Featherston, and S. Ahmed. “Deep Learning for Wound Tissue Segmentation: A Comprehensive Evaluation Using a Novel Dataset.” arXiv:2502.10652, 2025. Contribution: data creation, literature review, and implementation. [Preprint] [Repository]

4. Y. Huang, **Nidita Roy**, E. Dhar, U. Upadhyay, M. A. Kabir, M. Uddin, C.-L. Tseng, and S. Syed-Abdul. “Deep Learning Prediction Model for Patient Survival Outcomes in Palliative Care Using Actigraphy Data and Clinical Information.” *Cancers*, 15(8), 2232, 2023. [DOI]

5. **Nidita Roy** et al. “Enhancing Offensive Bengali Social Media Meme Detection: A Weighted Ensemble Architecture for Predicting Type and Target Classes.” 26th International Conference on Computer and Information Technology (ICCIT), 2023. Contribution: dataset creation and modeling. [IEEE Xplore]

Selected Projects

Diabetes Prediction	Applied ML
Machine-learning system for diabetes-risk prediction through a Flask-based web interface. [GitHub]	
Children Cloud	Web Application
Day-care booking and management system using PHP, HTML/CSS, Bootstrap, MySQL, and JavaScript. [GitHub]	
Chat-Site	Web Application
Browser-based chat application using PHP, MySQL, HTML, and CSS. [GitHub]	
CGPA Calculator	Web Utility
Web-based GPA calculation utility using PHP, HTML, and CSS. [GitHub]	
Recipe Website	Web Application
Recipe browsing and management website using HTML, CSS, Bootstrap, and JavaScript. [GitHub]	

Teaching Experience

Teaching Assistant, CSC 110, University of Victoria	Fall 2026 100 hours
Academic support for an introductory computer science course.	
Academic Assistant, Course Based Testing Facility, University of Victoria	Fall 2026 25 hours
Teaching Assistant, CSC 111: Fundamentals of Programming with Engineering Applications	Fall 2025–Spring 2026
• Supported 50+ undergraduate students; conducted recurring lab sections, graded labs and examinations, held office hours, and assisted with exam invigilation.	

Awards & Funding

Faculty of Graduate Studies Fellowship, University of Victoria	2026–2027 CAD \$20,000
Faculty of Graduate Studies Fellowship, University of Victoria	2025–2026

Skills

Research	Medical image analysis; class-incremental learning; continual learning; representation learning; multimodal healthcare AI; multiple-instance learning; image segmentation
ML & Programming	Python; PyTorch; TensorFlow; Keras; scikit-learn; OpenCV; Optuna; C; C++; Java
Research Computing	Linux; Git/GitHub; SLURM/HPC workflows; L ^A T _E X
Other	MySQL; Flask; HTML/CSS; JavaScript; Bootstrap; PrairieLearn

Leadership & Service

Vice President, CUET Computer Club

2023–2024

- Helped lead the national CUET Inter University Programming Contest and organized programming, machine-learning, and web-development events.
- Mentored participants in research methodology and feature engineering for machine-learning projects.
- [Leadership Evidence]

General Member, CUET Computer Club

2019–2023

Contributed to approximately 20 events, received a Best Contributor recognition, and mentored junior competitive programmers. [Best Contributor Certificate]

Microsoft Learn Student Ambassador

2021

Reached the Alpha milestone through community-building and technical-learning activities. [Credential]

Hacktoberfest Contributor

2021

Participated in open-source contribution activities during Hacktoberfest. [Evidence]

Selected Achievements & Training

- **2023: 8th place** in Sentiment Analysis of Code-Mixed Text [Evidence] and **7th place** in Multimodal Sentiment Analysis [Evidence], 2nd International Workshop on Computational Linguistics & Bangla Language Processing.
- **2021:** Ranked **1729 globally** in Google Code Jam I/O for Women.
- **2020: 9th place**, EDU Inter University Programming Contest [Evidence]; **Best Girl Programmer**, BdSON October CodeRace.
- **2019: 4th place**, CUET Intra University Programming Contest; **40th place**, National Girls' Programming Contest [Evidence].
- **2023:** Industrial attachment at **RedDot Digital Ltd.** involving image processing for tobacco-leaf quality classification. [Certificate]
- **2022:** Completed a 10-day practical training program in Artificial Intelligence and Machine Learning with Python. [Certificate]